

Abstract for the Tsukuba Science Edge 2026 [English form]

Your research stage is : ☐Planning ☐Preliminary ☒Result 【Required: Pick only one】

Area Check Most appropriate one 【Required】

☒Physics ☐Chemistry ☐Biology & Medical ☐Earth Sciences ☐Mathematics & Information

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Abstract of the Presentation

【Title】 Improving Wireless Charging Efficiency Using Photo-Sensing Multi-Transmitter Coil Arrays

【Outline and its First Inspiration】

The inspiration for this research stems from observing the gap in "Human-Computer Interaction" in daily life. In modern usage, users often move their bodies unconsciously while using mobile devices (e.g., tilting the device during gaming or operation). However, existing single-coil wireless charging technologies require high alignment precision. This leads to frequent power interruptions when "charging while using," which not only disrupts the user experience but also accelerates battery degradation due to repeated charging and discharging cycles.

Furthermore, we observed a serious energy waste issue in multi-coil charging pads currently on the market. Even if no device is present, or if a device occupies only a small area, the system often remains in a full-area standby or transmission state. This sparked a strong idea: "Can we design a system that accommodates dynamic human behavior (eliminating the need for precise alignment) while acting like biological neurons, releasing energy precisely only where it is needed?" Based on these pain points and inspirations, this study aims to develop an innovative system that integrates "blind-spot-free power supply" with "smart energy conservation."

【Novelty of Your Goal and its Original Value】

Current market solutions are polarized: cheap single-coil systems lack freedom, while expensive multi-coil systems face challenges regarding high cost and high standby power consumption. The novelty of this research lies in proposing a low-cost, high-efficiency "Hybrid Optimization Strategy":

1. Structural Optimization : Unlike simply increasing the number of coils, this study quantifies the magnetic coupling boundaries through pilot experiments. We propose a "Physical Overlap Matrix" layout that eliminates charging blind spots at the lowest possible cost.

2. Control Logic Innovation : We challenge traditional, complex electromagnetic sensing circuits by innovatively utilizing the physical characteristic of devices "blocking light." We employ low-cost photosensitive components (LDRs) for positioning. This not only significantly reduces system costs but also realizes a true "charge upon placement, stop upon removal" mechanism, fundamentally solving the energy waste problem of idle coils. This is an original solution that balances **User-friendliness** with **Eco-friendliness**.

【Creativity of Your Research Plan】

To achieve the above goals, our team designed a rigorous three-stage inquiry plan:

Stage 1 : Quantification of Physical Limits (Pilot Experiments)

We refused blind assembly and instead conducted a quantitative investigation of basic physical properties. Using the variable control method, we systematically tested the effects of signal frequency (50kHz-1000kHz), vertical distance between transmitter and receiver coils, and horizontal offset of both single and dual transmitter coils on the receiving

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voltage. We used an oscilloscope to build precise data models to provide theoretical support for subsequent designs.

Stage 2 : Development of Smart Control System

We utilized an Arduino UNO board combined with a photoresistor array to write the logic control program (as shown in Figure 9). The creativity of this stage lies in the "Dynamic Calibration" of the threshold. Instead of using a fixed value, we tested environmental background values to find the optimal parameter that distinguishes "object shielding" from background light, ensuring the system is not triggered falsely or unresponsive due to environmental changes (as shown in Figure 8).

Stage 3 : System Integration and Performance Verification

We integrated the optimized coil matrix with the smart control module. We also established an "Always-On Group" as a control group to practically measure and observe the final energy reception efficiency.

【Feasibility and its Prospect, based on Your Result and Interpretation】

This research has achieved significant preliminary results, confirming the feasibility of the design:

- 1. Establishment of Optimal Transmission Parameters :** Experimental data showed that while low frequencies have higher transmission efficiency, unstable fluctuations exist in the 50kHz-100kHz range. Balancing efficiency and stability, we finalized 150kHz (Figure 7) as the core transmission frequency and confirmed that effective magnetic coupling transmission is maintained within a vertical distance of 1cm (Figure 4).
- 2. Blind-Spot-Free Matrix Design :** Measurements revealed that when the horizontal offset of a single receiving coil relative to a single transmitting coil reaches 2cm (Figure 5), or when the offset between two transmitting coils reaches 4.5cm (Figure 6), the induced voltage drops below the usable 5V threshold. Based on this, we adopted a physical coil overlap of 0.5-1cm. Actual testing confirmed that this design successfully compensates for attenuation at the magnetic field boundaries, maintaining a stable voltage above 5V across the entire plane and thoroughly solving the disconnection issue caused by dynamic displacement.
- 3. High-Reliability Energy Saving Control :** Regarding photosensitive control, after repeated testing of environmental background values (approx. 120), we locked the trigger threshold at 50. Experiments proved that compared to an overly sensitive setting of 90 or a sluggish setting of 40, a setting of 50 accurately identifies device placement with a 100% success rate. It precisely activates the corresponding coil, significantly reducing the system's idle power consumption (as shown in Figures 14 and 15).

【Future Potential and Your Plan, if any】

We plan to upgrade our system using magnetic resonance to exceed the current 1cm vertical limit and employ flexible PCBs for curved surfaces. Our ultimate vision is to integrate this modular matrix into smart city infrastructure, such as sidewalks and benches, to eliminate "power anxiety" for mobile users. Aligning with SDGs 7 and 12, our design enhances charging efficiency and reduces battery wear, thereby contributing significantly to e-waste reduction and sustainable energy consumption.

【References】

1. Sample, A. P., Meyer, D. T., & Smith, J. R. (2011). *Analysis, design, and control of the dual-LCC magnetic resonance wireless charging system for electric vehicles*. *IEEE Transactions on Industrial Electronics*, 58(7), 2759–2768.
2. Kurs, A., Karalis, A., Moffatt, R., Joannopoulos, J. D., Fisher, P., & Soljačić, M. (2007). *Wireless power transfer via strongly coupled magnetic resonances*. *Science*, 317(5834), 83–86.
3. Chen, W., Liu, C., Lee, C. K., & Hui, S. Y. (2015). *A 2-D transmitting coil array for wireless power transfer using fixed beamforming*. *IEEE Transactions on Power Electronics*, 30(1), 222–234.

※You may fill this form **up to 2 printed pages** plus additional materials as below, but the latter will not be printed publication.

※**Your method and results** are required for reviewing so you must attach additional tables, charts, photos, etc., as many as needed. Please put respective number, title and caption. Also in the References list, all necessary references are numbered and referred in the text.

※For each description, please refer to the **Start-Up Check List**.

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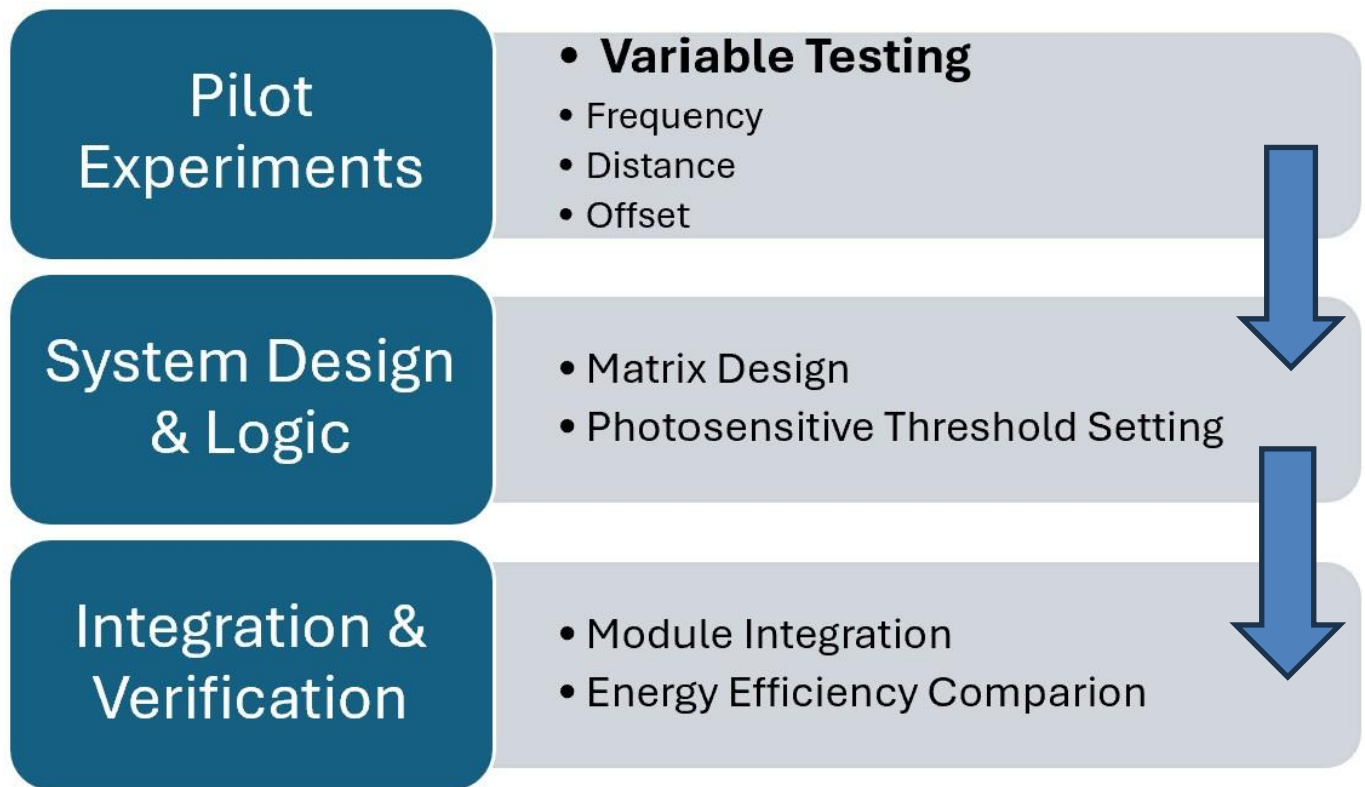


Figure 1: Research Methodology Flowchart

Figure Caption : This figure illustrates the three-stage process of how this research concretizes the scientific concept.

Stage 1 (Pilot Experiments) : Measuring the influence of various variables on the wireless charging receiving voltage.

Stage 2 (System Design & Logic Control) : Utilizing data from pilot experiments to design a suitable coil arrangement, and measuring the effect of different environmental light levels on photoresistor values to determine the optimal threshold.

Stage 3 (Integration & Verification) : Finally, integrating a model and comparing its energy reception performance.

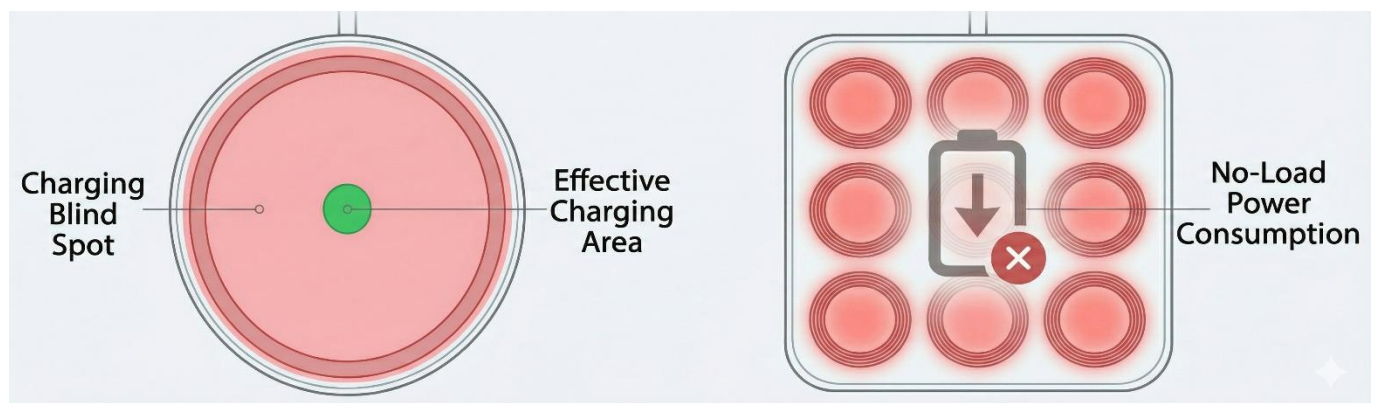


Figure 2 : Schematic diagram of the drawbacks of commercially available wireless charging pads

Figure Caption : The left image shows the drawback of single-coil wireless charging pads requiring precise alignment, while the right image shows the issue of traditional multi-coil charging pads being always-on and consuming power under no-load conditions.

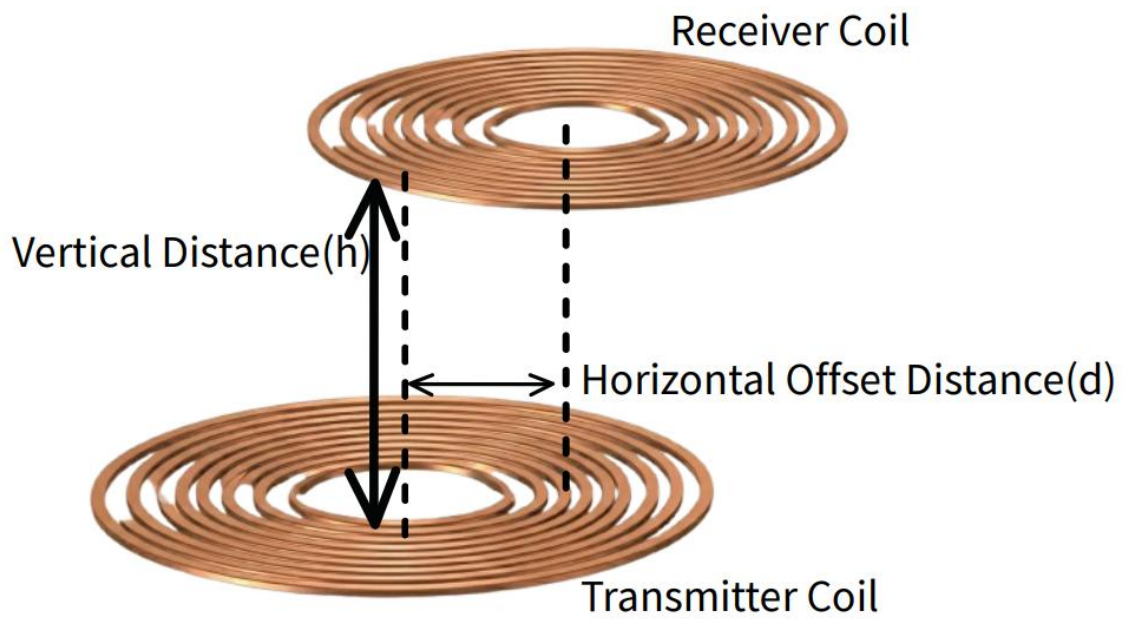


Figure 3 : Schematic diagram of coil horizontal offset and vertical displacement

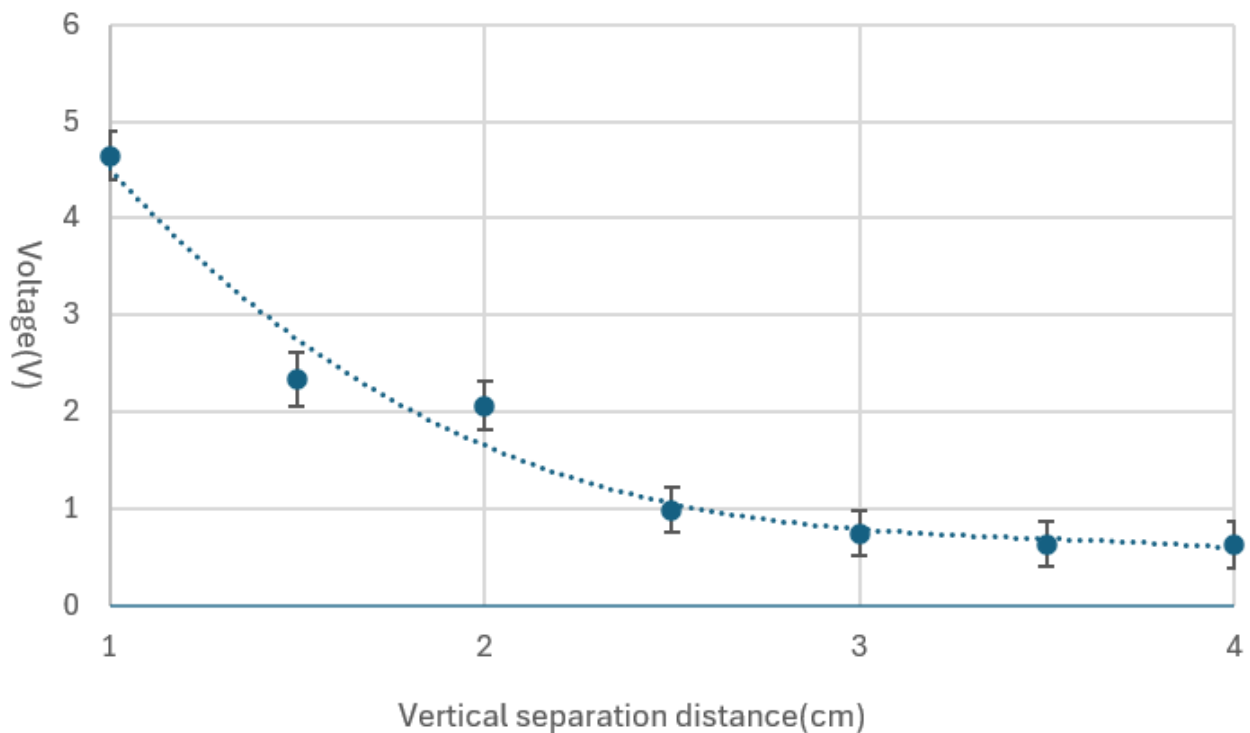


Figure 4 : The effect of vertical distance between transmitter and receiver coils on receiver voltage

Figure Caption : As shown in the figure above, the receiver voltage exhibits exponential decay as the vertical distance increases. At a distance of 1cm, the receiver voltage is already slightly below 5V; between 1~2cm, the voltage drops sharply; beyond 3cm, the induction intensity weakens significantly. Therefore, within 1cm is the optimal transmission distance.

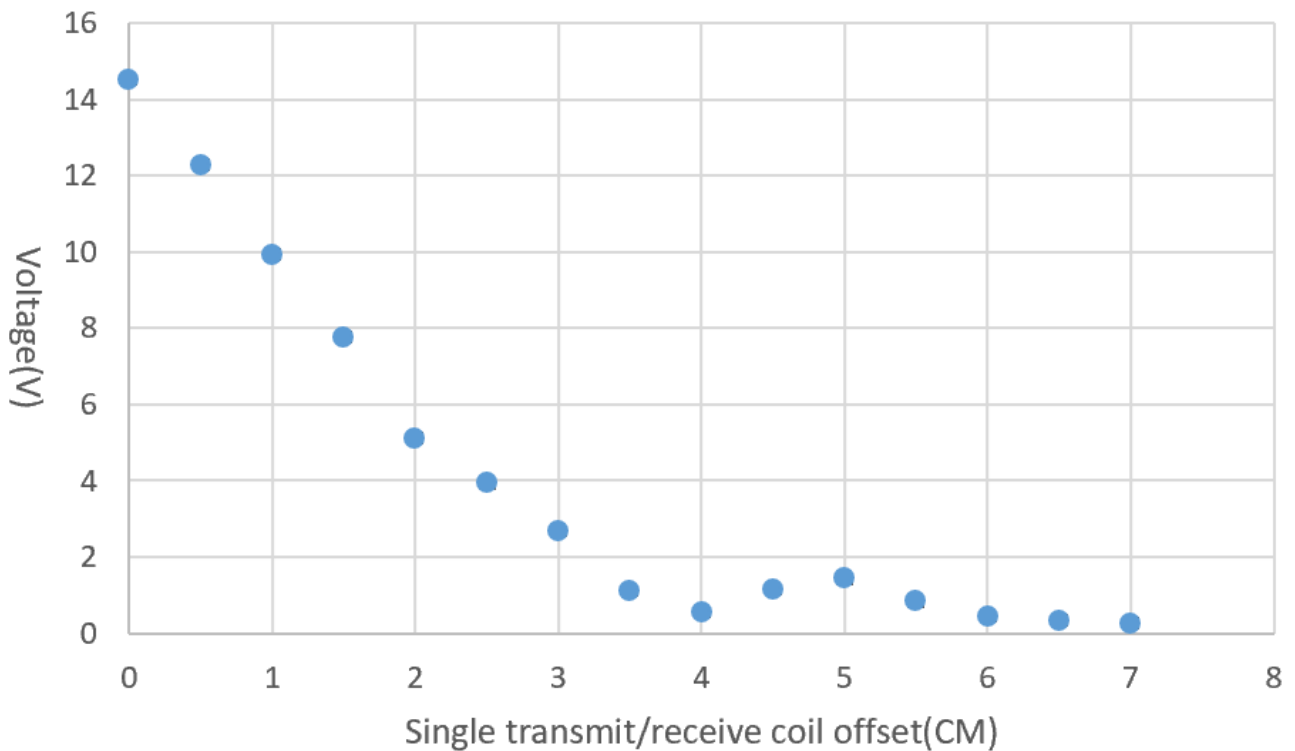


Figure 5 : The effect of horizontal offset distance between a single transmitter and receiver coil on receiver voltage

Figure Caption : This figure presents typical inductive coupling characteristics: as the horizontal displacement between the transmitting and receiving coils increases, the voltage decays rapidly. When the displacement exceeds 2cm, the receiver voltage drops to 5V; although there is a slight increase at 5cm displacement, the receiving voltage remains low.

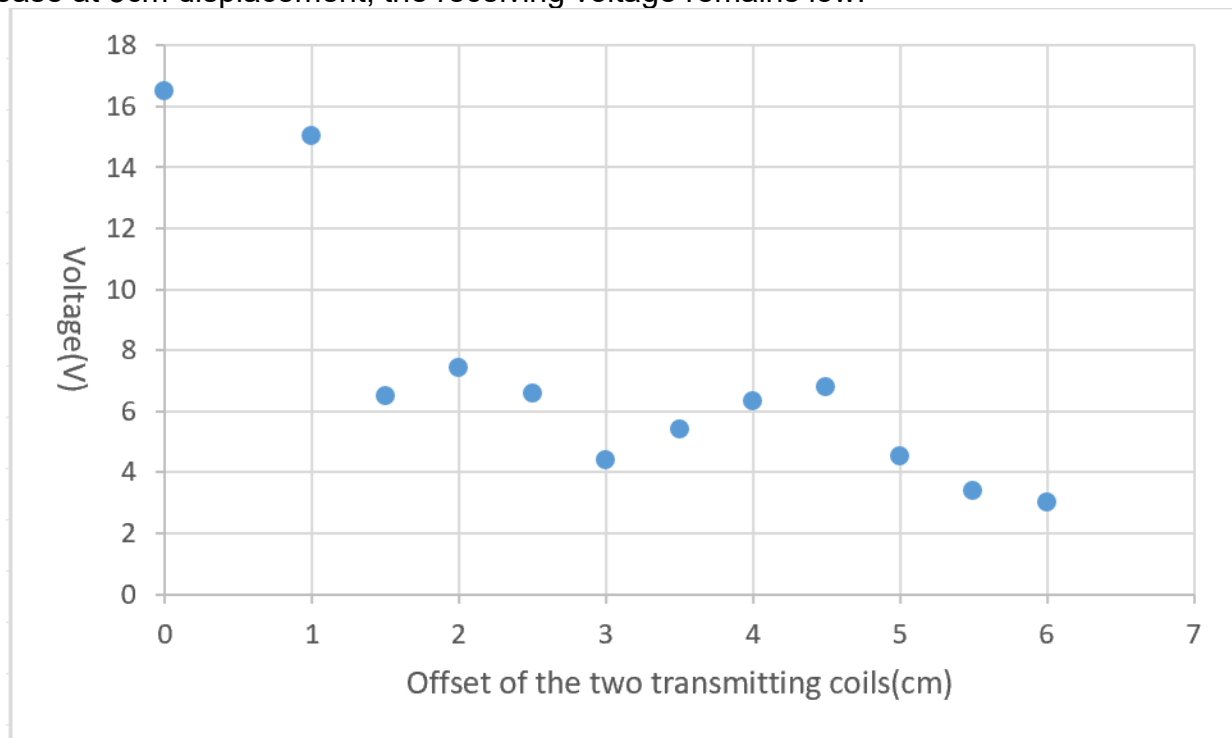


Figure 6 : The effect of horizontal offset distance between two transmitting coils on receiver voltage

Figure Caption : This figure shows that as the horizontal displacement between two transmitting coils increases, the induced voltage shows an overall downward trend. However, local rebounds occur near 2cm and 4.5cm, and between 1.5cm and 4.5cm, the receiver voltage is between 4~7V. After 5cm, it continues to decrease.

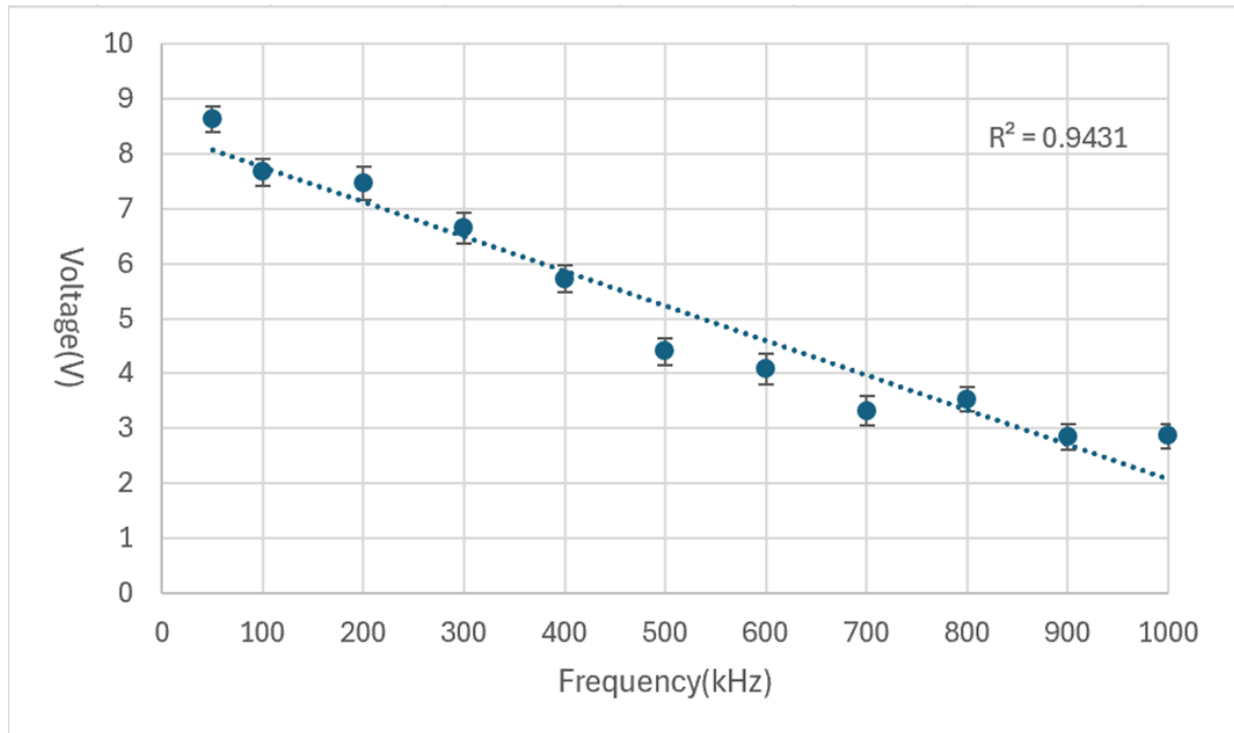


Figure 7 : The effect of transmitter signal frequency on receiver voltage

Figure Caption : Experimental data shows that the induced voltage decreases steadily as the operating frequency increases, with a linear fit correlation coefficient R^2 of 0.9431. The system voltage attenuation in high-frequency environments is highly predictable. Therefore, we chose the more stable 150kHz as the signal frequency.

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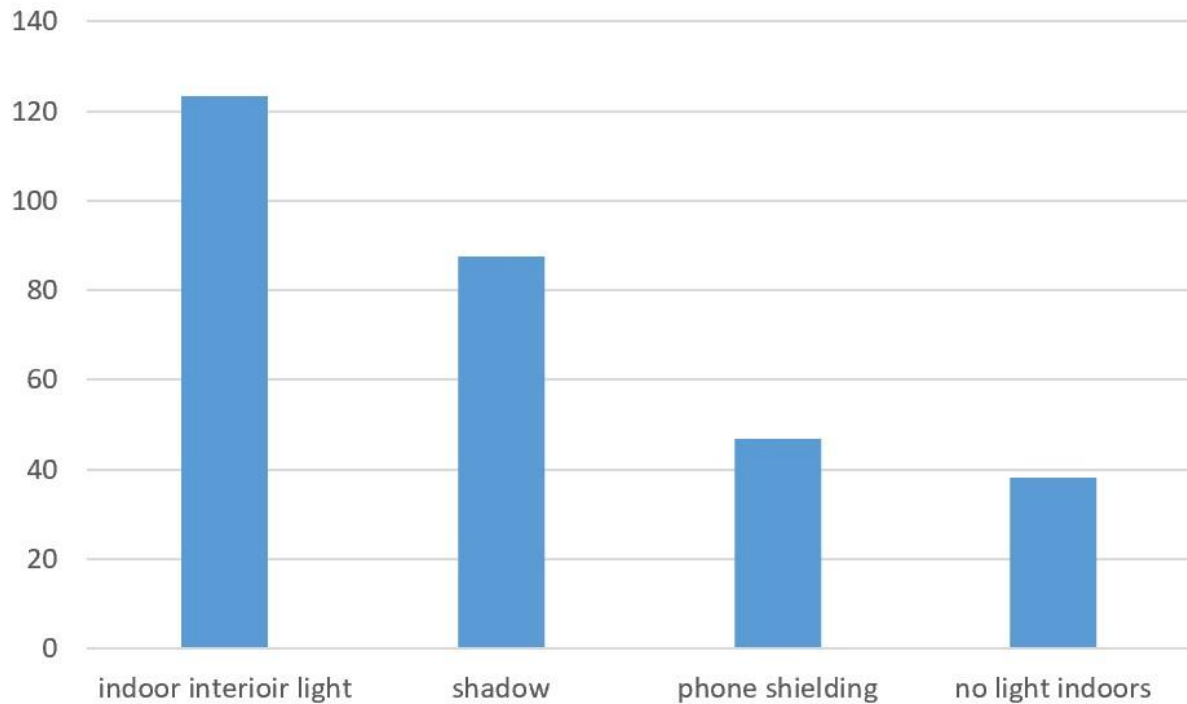


Figure 8 : Values displayed by the photoresistor under different lighting conditions

Figure Caption : As shown in the figure above, under normal indoor light, the average value is about 120; under shadow conditions, the average value is about 90; under phone shielding, the average value is about 40-50; in indoor no-light conditions, the average value is below 40. Therefore, to avoid the easy-to-mistrigger value of 80 and the hard-to-trigger value below 40, we selected the optimal threshold of 50.

```
const int LDR[] = {A0, A1, A2, A3, A4};
const int IN2 = 2; const int IN3 = 3; const int IN4 = 4;
const int threshold = 50;
void setup() {
  Serial.begin(9600);
  Serial.println("System Start: Monitoring LDR values...");
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);
  digitalWrite(IN2, HIGH);
  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, HIGH);}
void loop() {
  int v0 = analogRead(LDR[0]); int v1 = analogRead(LDR[1]); int v2 = analogRead(LDR[2]); int v3 = analogRead(LDR[3]); int v4 = analogRead(LDR[4]);
  Serial.print("A0:"); Serial.print(v0); Serial.print("\t");
  Serial.print("A1:"); Serial.print(v1); Serial.print("\t");
  Serial.print("A2:"); Serial.print(v2); Serial.print("\t");
  Serial.print("A3:"); Serial.print(v3); Serial.print("\t");
  Serial.print("A4:"); Serial.println(v4);
  if (v0 < threshold || v1 < threshold) digitalWrite(IN4, LOW);
  else digitalWrite(IN4, HIGH);
  if (v1 < threshold || v2 < threshold || v3 < threshold) digitalWrite(IN3, LOW);
  else digitalWrite(IN3, HIGH);
  if (v3 < threshold || v4 < threshold) digitalWrite(IN2, LOW);
  else digitalWrite(IN2, HIGH);
  delay(200); }
```

Figure 9 : Arduino UNO code to control a relay based on LDR values

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Figure Caption : The figure above shows the code for controlling the relay using photoresistor values.

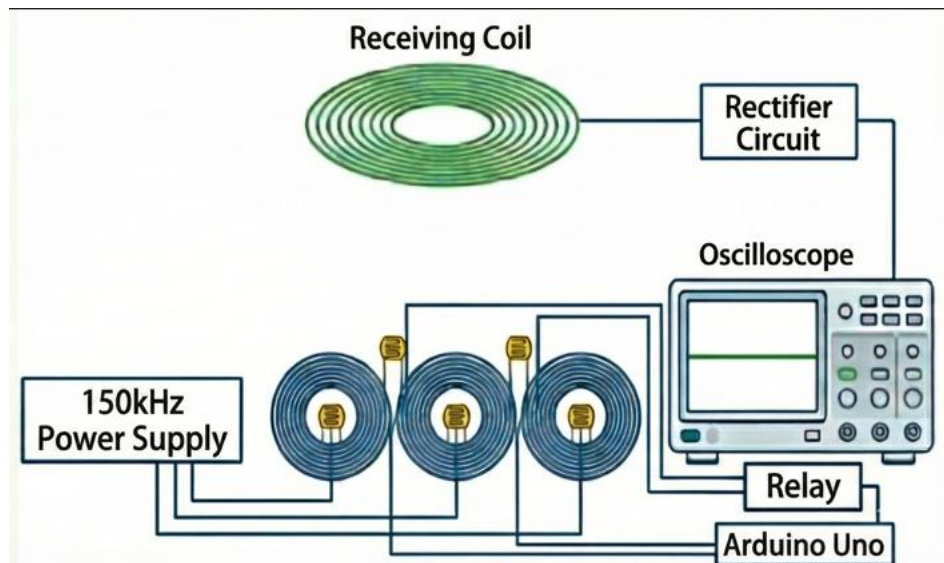


Figure 10 : Schematic diagram of the final product

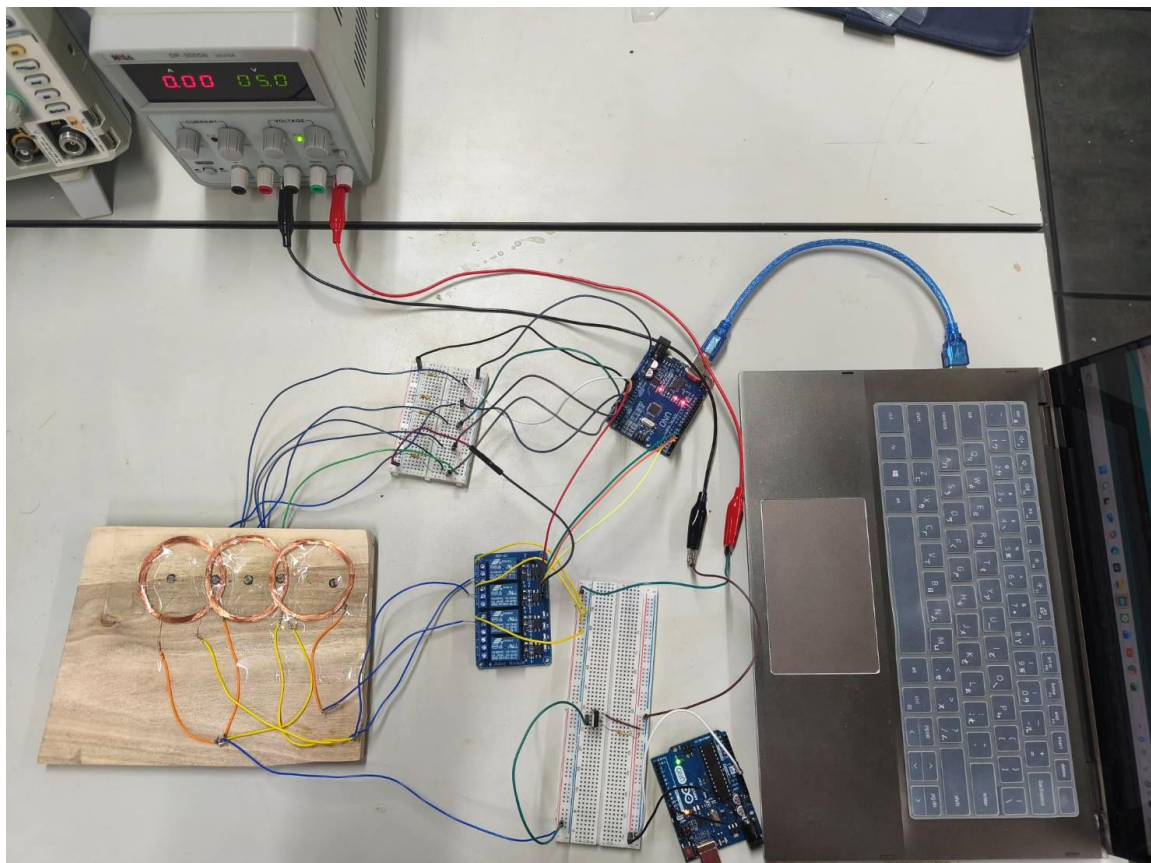


Figure 11 : Transmitter device diagram

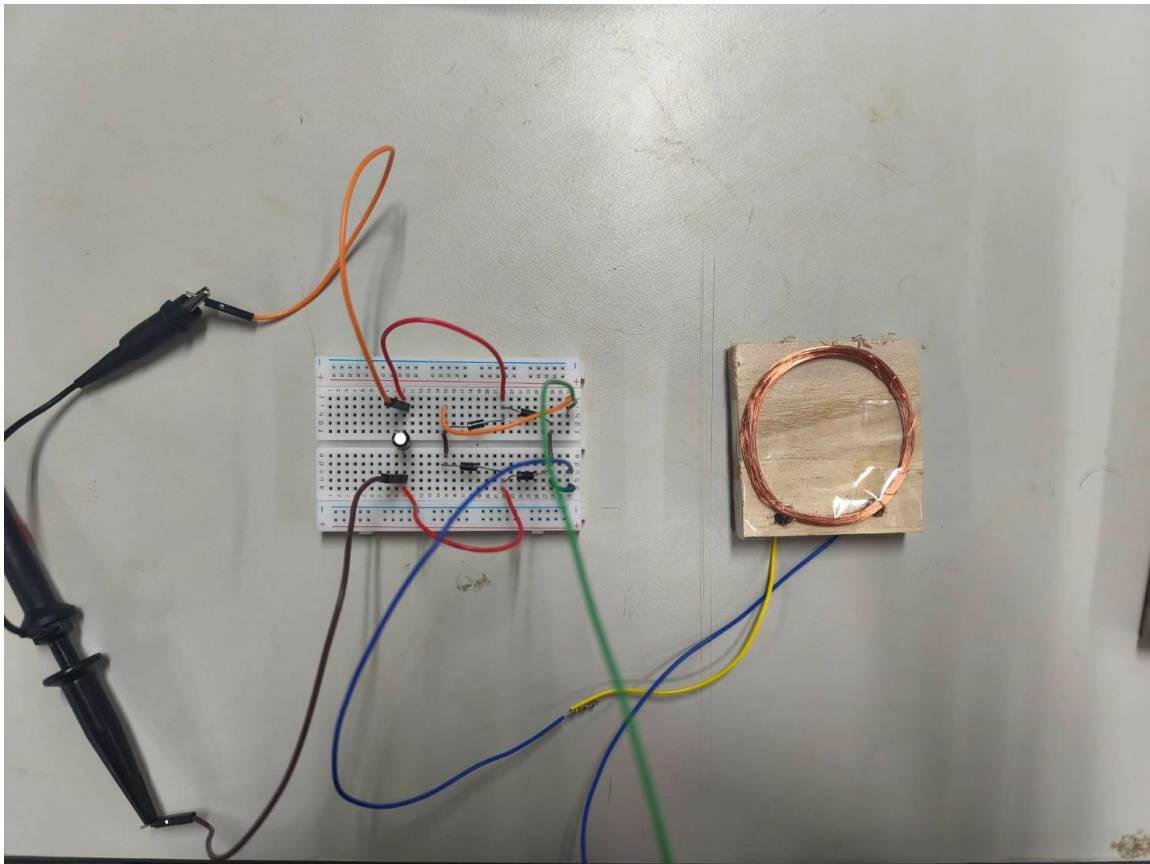


Figure 12 : Receiver device diagram

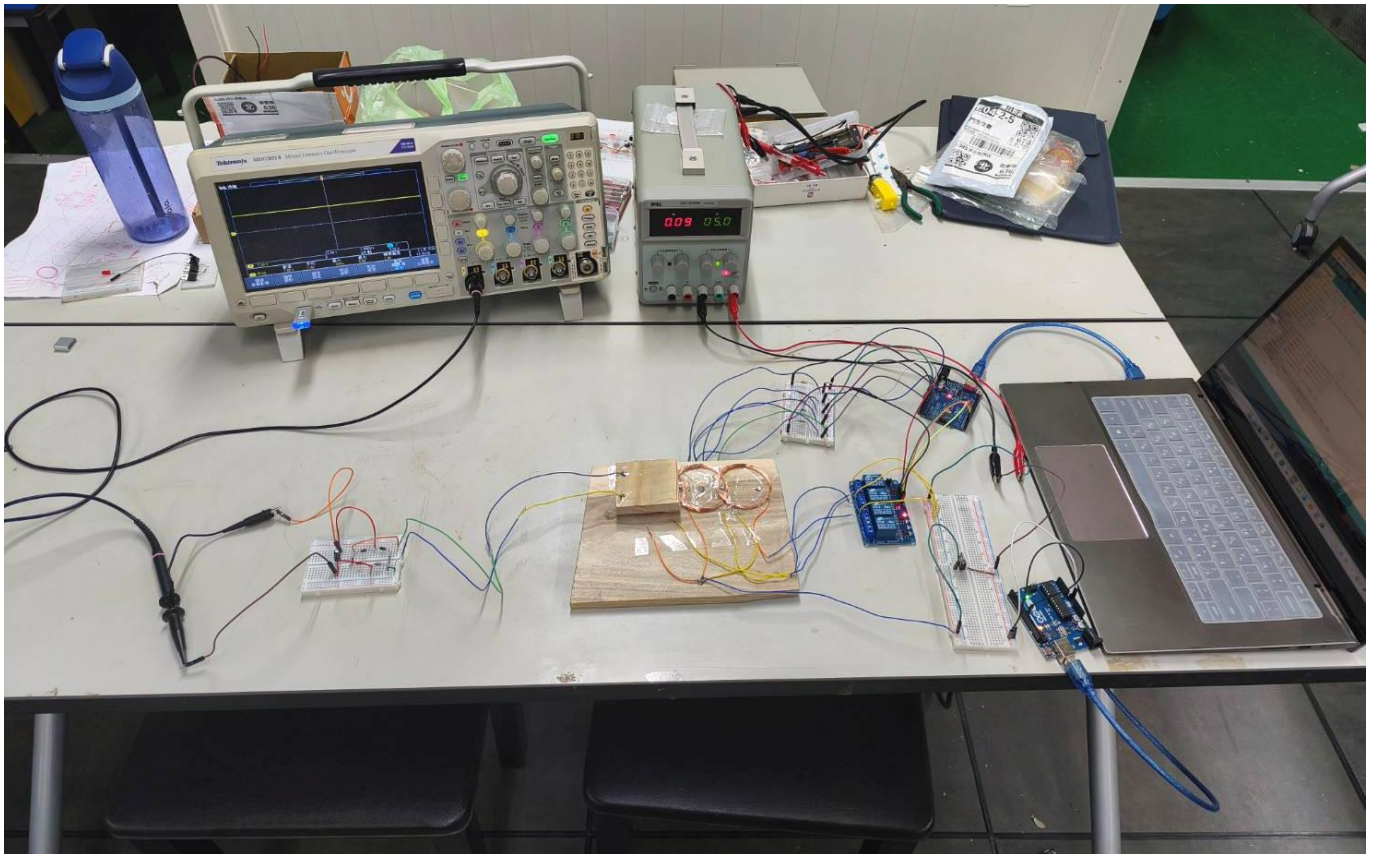


Figure 13 : Final experimental setup diagram

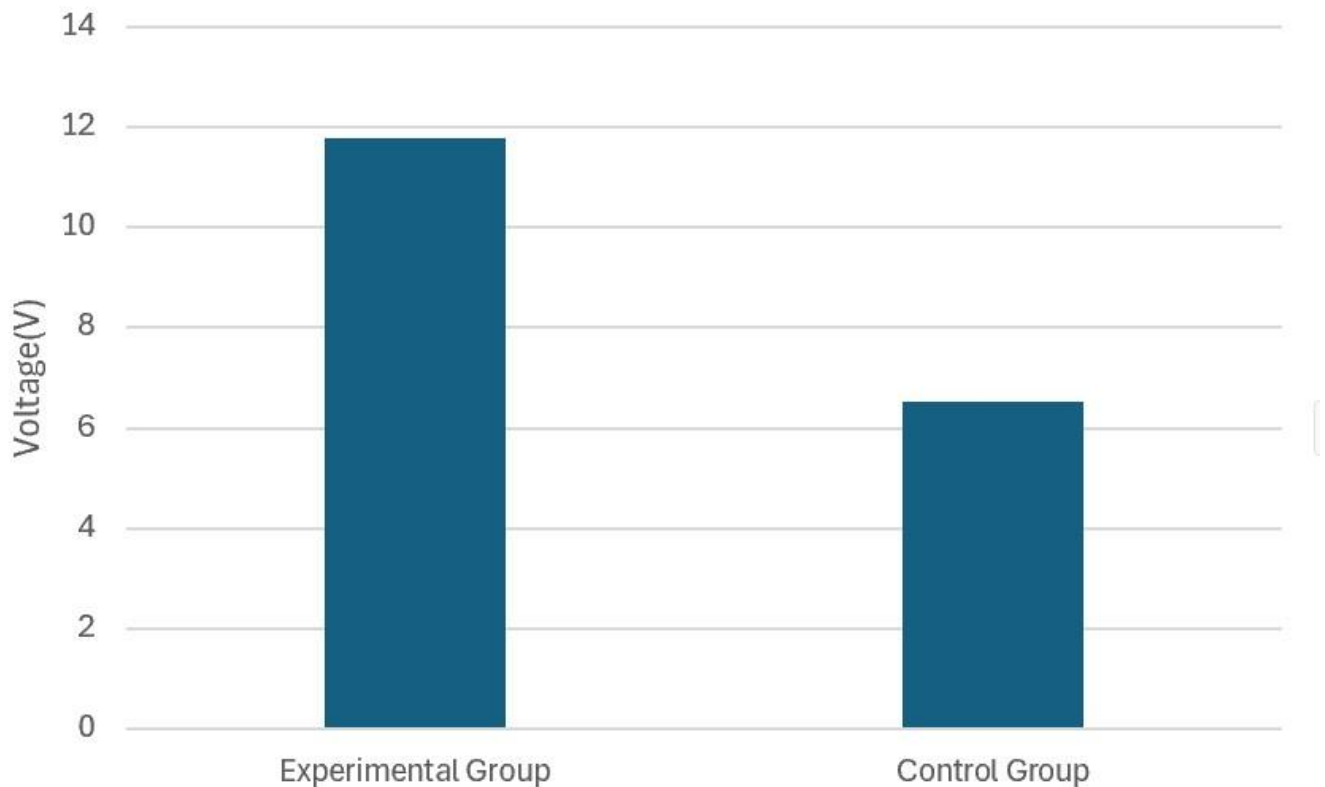


Figure 14 : Comparison chart of receiver voltage between the experimental group and the control group

Figure Caption : The experimental group includes photoresistor control to select coil switching, while the control group mimics a traditional multi-coil charging pad that is always on. Measuring the voltage at the receiving end reveals that the experimental group has a higher receiving voltage, implying that the energy transmitted by the experimental group is more concentrated.

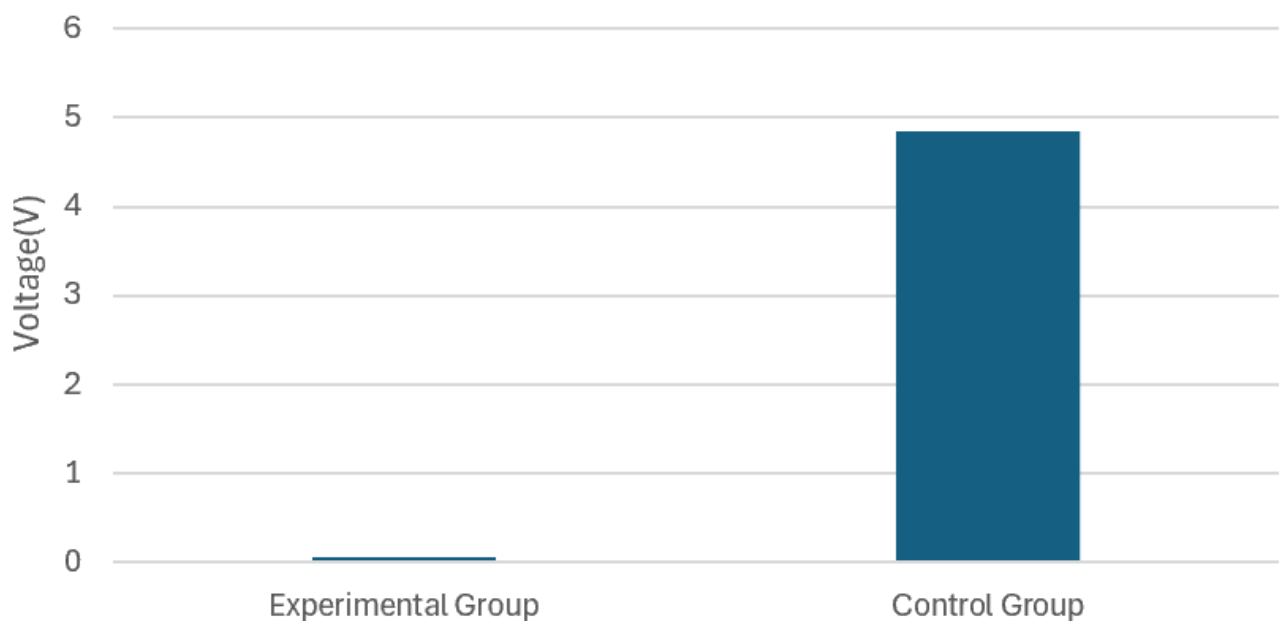


Figure 15 : Comparison of no-load energy consumption between the experimental group and the control group

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Figure Caption : The experimental group includes photoresistor control to select coil switching, while the control group mimics a traditional multi-coil charging pad that is always on. As seen in the figure, under no-load conditions, the transmitter voltage of the experimental group is close to 0, while the transmitter voltage of the control group is close to 5V. Thus, it can be determined that the experimental group has lower no-load energy consumption and is more power-efficient.